The Complex Arctangent

Key formulas and series expansions

Definition

$$\arctan z = \int_0^z \frac{1}{1+t^2}\,dt, \qquad z \neq \pm i$$

Branch points at $z = \pm i$.

Notes

Partial Fractions

$$1 + t^2 = (t-i)(t+i)$$

$$\frac{1}{1+t^2} = \frac{1}{2i}\left(\frac{1}{t-i} - \frac{1}{t+i}\right)$$

Notes

Logarithmic Form

$$\arctan z = \frac{1}{2i}\left[\log(t-i) - \log(t+i)\right]_0^z$$

$$\boxed{\arctan z = \frac{1}{2i}\log\left(\frac{i+z}{i-z}\right)}$$

Notes

Series About $z = 0$

For $|z| < 1$:

$$\arctan z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{2n+1}$$

Radius of convergence: $R = 1$.

Notes

Series About $z = 1$

Let $z = 1 + w$:

$$\arctan z = \frac{\pi}{4} + \frac{1}{2}w - \frac{1}{4}w^2 + \frac{1}{12}w^3 + \cdots$$

Radius: distance to $z = i$.

Notes